

**BISTRITA, Romania**WMO: **150850**Lat: **47.1492N**Lon: **24.5139E**Elev: **1205**StdP: **14.07**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 5.6        | 11.2 | 0.4                           | 5.9 | 9.1  | 5.6 | 7.7 | 13.2 | 20.3                  | 34.3 | 15.6 | 27.1 | 1.0                   | 50   | 0.457 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 22.9                   | 88.2            | 68.0 | 85.7 | 67.4 | 82.9 | 66.6 | 70.8                | 83.3 | 69.5 | 81.7 | 68.1 | 79.3 | 3.8                  | 290  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 67.0                            | 104.0 | 74.8 | 65.5 | 98.4 | 73.6 | 64.2 | 94.2 | 72.3 | 35.4          | 83.7 | 34.2 | 81.8 | 33.1 | 79.3 | 75.9              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |            | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |              |              |              |  |  |
|-------------------|-------------|------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|--------------|--------------|--------------|--|--|
|                   |             |            | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |              | n=50 years   |              |  |  |
| 1%                | 2.5%        | 5%         | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max          | Min          | Max          |  |  |
| (a)               | (b)         | (c)        | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)          | (o)          | (p)          |  |  |
| (4) <b>14.7</b>   | <b>10.9</b> | <b>8.4</b> | <b>-2.0</b>                | <b>92.6</b> | <b>6.9</b>         | <b>4.8</b> | <b>-7.0</b>                                        | <b>96.0</b> | <b>-11.0</b> | <b>98.9</b> | <b>-14.9</b> | <b>101.6</b> | <b>-19.9</b> | <b>105.1</b> |  |  |
| (5)               |             |            | <b>-2.3</b>                | <b>73.9</b> | <b>6.8</b>         | <b>1.4</b> | <b>-7.2</b>                                        | <b>75.0</b> | <b>-11.2</b> | <b>75.8</b> | <b>-15.1</b> | <b>76.6</b>  | <b>-20.0</b> | <b>77.6</b>  |  |  |

**Monthly Climatic Design Conditions**

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 49.7   | 28.3  | 33.4  | 40.0  | 50.8  | 59.3  | 66.1  | 68.1  | 68.6  | 60.1  | 50.0  | 40.6  | 30.4  |      |
|                                                                                  | DBStd                                                                            | 16.00  | 8.06  | 7.76  | 7.31  | 7.97  | 6.16  | 5.59  | 5.17  | 6.04  | 7.86  | 7.29  | 8.23  | 8.46  |      |
|                                                                                  | HDD50                                                                            | 2557   | 673   | 466   | 318   | 87    | 6     | 1     | 0     | 0     | 8     | 95    | 296   | 607   |      |
|                                                                                  | HDD65                                                                            | 5996   | 1138  | 886   | 776   | 428   | 192   | 52    | 32    | 34    | 188   | 466   | 732   | 1072  |      |
|                                                                                  | CDD50                                                                            | 2448   | 0     | 0     | 7     | 109   | 295   | 483   | 561   | 575   | 310   | 94    | 14    | 0     |      |
|                                                                                  | CDD65                                                                            | 414    | 0     | 0     | 0     | 1     | 16    | 85    | 128   | 144   | 40    | 0     | 0     | 0     |      |
|                                                                                  | CDH74                                                                            | 4295   | 0     | 0     | 0     | 38    | 230   | 827   | 1211  | 1514  | 465   | 10    | 0     | 0     |      |
|                                                                                  | CDH80                                                                            | 1292   | 0     | 0     | 0     | 2     | 28    | 208   | 338   | 585   | 131   | 0     | 0     | 0     |      |
| Wind                                                                             | WSAvg                                                                            | 2.7    | 1.9   | 2.3   | 3.4   | 3.7   | 3.3   | 3.0   | 3.0   | 2.9   | 2.7   | 2.3   | 2.1   | 2.1   |      |
| Precipitation                                                                    | PrecAvg                                                                          | 27.7   | 1.8   | 1.4   | 1.5   | 2.1   | 3.0   | 3.7   | 3.5   | 2.6   | 2.1   | 1.9   | 1.9   | 2.0   |      |
|                                                                                  | PrecMax                                                                          | 38.4   | 4.5   | 3.7   | 4.4   | 4.4   | 8.5   | 7.5   | 7.7   | 7.5   | 6.0   | 4.8   | 4.3   | 5.0   |      |
|                                                                                  | PrecMin                                                                          | 16.5   | 0.3   | 0.0   | 0.1   | 0.4   | 0.6   | 0.5   | 0.6   | 0.3   | 0.1   | 0.0   | 0.0   | 0.0   |      |
|                                                                                  | PrecStd                                                                          | 5.0    | 1.0   | 0.8   | 1.0   | 0.9   | 1.6   | 1.5   | 1.6   | 1.7   | 1.4   | 1.3   | 1.0   | 1.1   |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 48.0  | 58.1  | 67.3  | 78.1  | 83.2  | 88.4  | 89.9  | 93.7  | 88.4  | 75.3  | 66.4  | 50.0  |      |
|                                                                                  |                                                                                  | MCWB   | 42.0  | 47.9  | 51.6  | 58.2  | 64.1  | 69.9  | 69.6  | 69.3  | 65.0  | 59.8  | 54.9  | 45.6  |      |
|                                                                                  | 2%                                                                               | DB     | 43.6  | 51.9  | 61.2  | 74.1  | 79.7  | 85.4  | 86.7  | 89.6  | 83.9  | 72.2  | 61.2  | 45.4  |      |
|                                                                                  |                                                                                  | MCWB   | 40.1  | 44.9  | 49.1  | 57.0  | 63.4  | 68.7  | 68.6  | 67.2  | 63.9  | 59.1  | 52.5  | 42.8  |      |
|                                                                                  | 5%                                                                               | DB     | 40.4  | 47.3  | 55.7  | 69.5  | 76.4  | 82.2  | 83.8  | 86.3  | 79.5  | 67.6  | 56.6  | 42.2  |      |
|                                                                                  |                                                                                  | MCWB   | 37.9  | 42.7  | 46.1  | 55.2  | 62.1  | 67.5  | 67.4  | 67.3  | 62.6  | 57.6  | 50.5  | 40.6  |      |
|                                                                                  | 10%                                                                              | DB     | 37.4  | 43.8  | 51.6  | 65.4  | 72.8  | 78.9  | 81.0  | 83.0  | 74.8  | 62.7  | 52.5  | 39.9  |      |
|                                                                                  |                                                                                  | MCWB   | 35.7  | 40.2  | 44.0  | 53.6  | 60.7  | 66.2  | 67.3  | 66.5  | 62.0  | 55.5  | 48.4  | 38.8  |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 43.9  | 48.5  | 53.5  | 60.5  | 67.5  | 72.8  | 72.3  | 72.5  | 68.3  | 63.2  | 55.7  | 45.9 |
|                                                                                  |                                                                                  |        | MCDB  | 46.2  | 56.0  | 65.5  | 71.6  | 77.3  | 85.2  | 83.4  | 86.5  | 83.4  | 69.6  | 62.2  | 48.2 |
| 2%                                                                               |                                                                                  | WB     | 40.6  | 45.7  | 49.8  | 58.3  | 65.1  | 70.1  | 70.8  | 70.4  | 66.1  | 60.5  | 53.4  | 43.2  |      |
|                                                                                  |                                                                                  | MCDB   | 43.4  | 50.8  | 58.9  | 70.8  | 75.6  | 82.0  | 82.4  | 84.5  | 78.0  | 69.1  | 59.0  | 44.9  |      |
| 5%                                                                               |                                                                                  | WB     | 38.2  | 43.2  | 47.5  | 56.3  | 63.4  | 68.5  | 69.5  | 68.9  | 64.2  | 58.3  | 51.3  | 40.9  |      |
|                                                                                  |                                                                                  | MCDB   | 40.1  | 47.0  | 54.6  | 67.4  | 73.6  | 79.5  | 80.7  | 82.1  | 74.7  | 66.6  | 56.3  | 42.2  |      |
| 10%                                                                              |                                                                                  | WB     | 36.0  | 40.7  | 45.0  | 54.5  | 61.8  | 67.1  | 68.1  | 67.3  | 62.5  | 55.9  | 48.6  | 38.9  |      |
|                                                                                  |                                                                                  | MCDB   | 37.0  | 43.2  | 50.2  | 63.5  | 71.2  | 76.8  | 78.0  | 79.3  | 72.2  | 61.8  | 52.0  | 39.9  |      |
| Mean Daily<br>Temperature<br>Range                                               |                                                                                  | 5% DB  | MDBR  | 12.1  | 15.1  | 18.0  | 20.6  | 20.9  | 21.1  | 21.6  | 22.9  | 20.8  | 19.6  | 14.5  | 10.3 |
|                                                                                  |                                                                                  |        | MCDBR | 14.5  | 20.0  | 25.5  | 28.2  | 26.6  | 25.8  | 26.3  | 28.3  | 28.0  | 27.8  | 22.0  | 13.0 |
|                                                                                  | MCWBR                                                                            |        | 10.7  | 13.1  | 14.5  | 13.2  | 12.4  | 11.3  | 10.7  | 9.8   | 10.7  | 15.4  | 13.3  | 10.1  |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 12.5  | 18.4  | 22.9  | 25.0  | 24.2  | 23.8  | 23.7  | 25.2  | 24.5  | 24.7  | 20.0  | 12.1  |      |
|                                                                                  |                                                                                  | MCWBR  | 9.4   | 12.7  | 13.7  | 12.7  | 12.2  | 11.5  | 10.8  | 10.4  | 10.8  | 14.4  | 12.5  | 9.7   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             |        | 0.306 | 0.335 | 0.378 | 0.430 | 0.414 | 0.410 | 0.431 | 0.434 | 0.394 | 0.361 | 0.335 | 0.308 |      |
|                                                                                  | taud                                                                             |        | 2.312 | 2.256 | 2.219 | 2.135 | 2.251 | 2.293 | 2.240 | 2.230 | 2.306 | 2.382 | 2.393 | 2.362 |      |
|                                                                                  | Ebn at Noon                                                                      |        | 244   | 259   | 265   | 260   | 270   | 272   | 263   | 257   | 257   | 248   | 232   | 229   |      |
|                                                                                  | Edh at Noon                                                                      |        | 24    | 31    | 38    | 45    | 42    | 41    | 42    | 41    | 35    | 28    | 22    | 21    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           |        | 390   | 655   | 1002  | 1340  | 1647  | 1761  | 1750  | 1616  | 1154  | 785   | 443   | 299   |      |
|                                                                                  | RadStd                                                                           |        | 56    | 75    | 127   | 134   | 163   | 147   | 160   | 148   | 143   | 104   | 61    | 40    |      |